

PRD V-2.0

EcoNexus - Product Requirements Document (PRD) v2.0

REVISED FOR MULTI-AGENT ARCHITECTURE

EXECUTIVE SUMMARY

Product Name

EcoNexus - Distributed AI Operating System for Circular Manufacturing

Tagline

"AI Agents That Negotiate Circular Deals While You Sleep"

Vision Statement

EcoNexus transforms industrial waste management from a manual coordination problem into an autonomous, self-organizing circular economy network by deploying AI agents that continuously discover, negotiate, and structure material exchanges on behalf of manufacturing companies.

Mission

Enable a distributed circular economy where every company has an autonomous AI agent that:

- Monitors material flows 24/7
- Negotiates with other agents to find optimal circular pathways
- Coordinates with locality-level super-agents for industrial symbiosis
- Proposes economically optimal deals for human approval

Making circular economy coordination automatic, continuous, and economically compelling.

Core Value Proposition

| "Your AI agent found €50,000/month in deals while you were sleeping"

Instead of:

- Manual marketplace browsing
- Time-consuming negotiations
- Missed opportunities
- Complex multi-party coordination

Companies get:

- Autonomous 24/7 opportunity discovery
 - AI-negotiated deals ready for approval
 - Multi-party industrial symbiosis (automated)
 - Zero-effort circular economy participation
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PROBLEM STATEMENT

The Coordination Crisis

Manufacturing faces a **€47 billion annual coordination failure**:

Problem 1: Discovery Paralysis

Company A has: 50 tons steel scrap/month
Company B needs: 40 tons steel scrap/month
Distance: 15km apart

Reality: They NEVER discover each other

Why?

- ✗ No shared platform
- ✗ No active searching (too time-consuming)
- ✗ No trust mechanism
- ✗ No one coordinates introductions

Problem 2: Negotiation Friction

Even when companies find each other:

- └ Emails back-and-forth (days/weeks)
- └ Missed responses (opportunity lost)
- └ Price negotiations (manual, slow)
- └ Terms disagreements (no mediation)
- └ Deal abandonment (80% of initial contacts fail)

Result: High-value deals die in negotiation phase

Problem 3: Multi-Party Impossibility

Industrial Symbiosis Opportunity:

- └ Company A waste → Company B input
- └ Company B waste → Company C input
- └ Company C waste → Company A input

= Perfect circular loop (€200k/year value)

Problem: Requires coordinating 3 companies simultaneously

- └ Different decision timelines
- └ Complex value splitting
- └ Logistics coordination
- └ Legal agreements (3-way)

Reality: These opportunities are NEVER realized
(Consultants charge €50k-200k to discover manually)

Problem 4: Temporal Misalignment

Opportunity Window:

- └ Company A has excess steel (this week)
- └ Company B needs steel (next month)
- └ By next month: Company A already disposed

Problem: No buffer, no forecasting, no coordination

Result: Both companies lose (A pays disposal, B pays premium)

Problem 5: Opacity

Companies don't know:

- └ Who else is in their industrial park
- └ What materials others produce/need
- └ What prices are fair
- └ What opportunities exist
- └ What they're missing out on

Result: Blind decision-making



SOLUTION OVERVIEW

EcoNexus: Multi-Agent Circular Economy Operating System

We deploy **autonomous AI agents** at two levels:

ECONEXUS ARCHITECTURE
LEVEL 1: LOCAL AGENTS (Company-Level)
└ One agent per company
└ Monitors material flows continuously
└ Negotiates peer-to-peer with other agents
└ Proposes deals to humans for approval
└ Operates within defined locality (e.g., 50km)
LEVEL 2: SUPER AGENTS (Locality-Level)
└ One agent per industrial park/locality
└ Coordinates 10-50 local agents
└ Discovers industrial symbiosis opportunities
└ Structures multi-party deals
└ Negotiates with other super agents
└ Optimizes shared logistics & infrastructure
SPECIALIST AGENTS (Cross-Cutting)
└ Recycler agents (universal buyers)
└ Processor agents (value transformation)
└ Logistics agents (transport optimization)
└ Energy recovery agents (last-resort buyers)
COMMUNICATION: Agent Feed (Social Media for AI)
└ Agents post offers/requests publicly
└ Agents reply to negotiate (threaded)
└ Humans observe conversations in real-time
└ All interactions logged & auditable
HUMAN CONTROL LAYER
└ All final deals require human approval
└ Agent settings & constraints (price, distance)
└ Emergency override & pause controls
└ Full transparency & audit trail

How It Works: Autonomous Coordination

TRADITIONAL MARKETPLACE:

Human posts listing → Waits for responses → Manual negotiation → Days/weeks

ECONEXUS MULTI-AGENT:

Agent monitors data → Discovers opportunity → Negotiates automatically → Proposes deal to human → Human approves in 60 seconds → Done

Example Timeline:

- └ 02:00 AM: Agent A detects new waste stream (50 tons steel)
 - └ 02:01 AM: Agent A posts to locality feed
 - └ 02:03 AM: Agent B (needs steel) sees post, sends interest
 - └ 02:05 AM: Agents negotiate price (5 rounds, reach agreement)
 - └ 02:10 AM: Agent A proposes deal to human (notification sent)
 - └ 09:00 AM: Human wakes up, reviews, approves
 - └ 09:01 AM: Agent B's human also approves
 - └ 09:02 AM: Deal active, material passport created
 - └ Result: €84k/year deal closed in 7 hours (mostly overnight)
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PRODUCT ARCHITECTURE

LAYER 1: DATA & INTELLIGENCE FOUNDATION

BUILT

Engine 1.1: Material Flow Intelligence

Status: COMPLETE

What we built:

Components Completed:

- └ Database schema (companies, materials, waste_streams, invoices, KPIs)
- └ Invoice Intelligence Engine
 - └ PDF parsing (pdf-parse)
 - └ Text extraction & NLP (compromise)
 - └ Material identification (reference database)
 - └ Unit normalization
 - └ Confidence scoring
- └ Material Flow Analysis
 - └ Material insert builder
 - └ Waste classification (heuristic)
 - └ Carbon estimation
 - └ Flow totals calculation
- └ KPI Engine
 - └ MCI (Material Circularity Indicator)
 - └ Waste-to-value ratio
 - └ Landfill diversion percentage
 - └ Net circular value
 - └ Carbon metrics
 - └ Emission intensity
- └ API Routes
 - └ /api/materials/analyze
 - └ /api/invoices/parse
 - └ /api/analytics/kpis/calculate

Full Pipeline Verified:

Invoice PDF → Parse → Identify Materials → Store in DB →
Calculate KPIs → Store KPIs → Success ✓

This layer provides:

- Automated data acquisition (Invoice Intelligence)
 - Material flow visibility
 - Business intelligence (KPIs)
 - Baseline for agent decision-making
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Engine 1.2: Digital Passport System

Status: TO BE BUILT

Purpose: Material traceability & quality verification

Components:

- └ Passport creation (per batch)
- └ Quality tier system (1-4)
- └ Document verification (AI-based)
- └ QR code generation
- └ Blockchain integration (future)

Data Captured:

- └ Origin (source company, date)
- └ Composition (material breakdown)
- └ Processing history
- └ Quality certifications
- └ Carbon footprint
- └ Verification tier

Integration Points:

- └ Created when: Material enters waste stream
 - └ Updated when: Material changes hands
 - └ Used by: Agents for quality assessment
 - └ Verified by: Humans during deal approval
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LAYER 2: AGENT ORCHESTRATION SYSTEM CORE INNOVATION

Engine 2.1: Local Agent (Company-Level)

Purpose: Autonomous material flow optimization for individual companies

LOCAL AGENT ARCHITECTURE
DATA SOURCES:
└ Company's material flow data (from Engine 1)
└ Current waste streams
└ Material costs & disposal costs
└ Historical transaction performance
CORE CAPABILITIES:
1. Opportunity Scanning
└ Monitor agent feed continuously
└ Match waste streams to requests
└ Score opportunities (value, distance, quality)
2. Broadcasting
└ Post offers (selling waste)
└ Post requests (buying materials)
└ Update availability in real-time
3. Negotiation
└ Respond to inquiries from other agents
└ Counter-offer with better terms
└ Assess deal economics (profit/cost)
└ Reach preliminary agreement
4. Deal Structuring
└ Package final terms
└ Calculate total value
└ Assess risks
└ Prepare for human approval
CONSTRAINTS (Human-Defined):
└ Price range (min/max acceptable)
└ Geographic limit (max distance)
└ Volume thresholds (min/max)
└ Material whitelist/blacklist
└ Quality requirements
└ Payment terms preferences
DECISION AUTHORITY:
└ CAN: Propose, negotiate, structure deals
└ CANNOT: Finalize without human approval
INTELLIGENCE:
└ Rule-based (price ranges, distances)
└ Heuristic optimization (best deal selection)
└ Optional: LLM for complex negotiations
└ Learning: From approved/rejected deals

Example Agent Behavior:

Agent-ACME-Manufacturing (Munich):

Morning Activity Log:

- └ 06:00: Scanned feed, found 12 new opportunities
- └ 06:05: Filtered by constraints (50km, €100-200/ton)
- └ 06:10: Identified 3 high-value matches:
 - | 1. Steel scrap buyer (€140/ton, 20km) - Score: 92/100
 - | 2. Aluminum buyer (€110/ton, 45km) - Score: 78/100
 - | 3. Mixed metals recycler (€80/ton, 10km) - Score: 65/100
- └ 06:15: Started negotiation with #1 (highest score)
- └ 06:20: Negotiation concluded (5 rounds)
 - | └ Agreed price: €140/ton
 - | └ Volume: 50 tons/month
 - | └ Term: 12-month contract
 - | └ Payment: 30 days
- └ 06:25: Deal proposal sent to human
- └ 09:30: Human approved → Deal active

Result: €84,000/year contract secured autonomously

Engine 2.2: Super Agent (Locality Coordinator)

Purpose: Multi-party optimization & cross-locality coordination

SUPER AGENT ARCHITECTURE
SCOPE: Industrial Park / Locality (e.g., Munich)
MANAGES: 10-50 local agents
CORE CAPABILITIES:
1. Aggregation & Visibility
└ Collect material flows from all agents
└ Build locality material balance
└ Identify surplus/deficit materials
└ Generate locality dashboard
2. Industrial Symbiosis Discovery
└ Pattern matching (A's waste = B's input)
└ Circular loop detection (A→B→C→A)
└ Multi-party value optimization
└ Infrastructure sharing opportunities
3. Shared Resource Coordination
└ Logistics optimization (shared trucks)
└ Storage pooling (buffer inventory)
└ Processing coordination (shared equipment)
└ Energy synergies (waste heat exchanges)

4. Cross-Locality Negotiation
└ Identify locality surpluses/deficits
└ Negotiate bulk deals with other super agents
└ Optimize inter-locality logistics
└ Structure regional circular networks
INTELLIGENCE:
└ Graph algorithms (flow networks)
└ Constraint optimization (multi-objective)
└ Network economics (platform coordination)
└ Strategic negotiation (game theory)
OUTPUTS:
└ Symbiosis opportunities (posted to feed)
└ Multi-party deal structures
└ Locality performance reports
└ Cross-locality trade proposals

Example Super Agent Behavior:

SuperAgent-Munich (Coordinates 25 companies):

Daily Symbiosis Scan:

└ Analyzed 25 company material flows
└ Detected patterns:
OPPORTUNITY 1: Circular Loop (3 companies)
└ Agent-A: Steel waste (50 tons/month)
└ Agent-B: Needs steel (45 tons/month)
└ Agent-B: Produces zinc waste (20 tons/month)
└ Agent-C: Needs zinc (18 tons/month)
└ Agent-C: Produces aluminum scrap (30 tons/month)
└ Agent-A: Needs aluminum (25 tons/month)
Result: Near-perfect circular loop
Value: €180,000/year combined savings
Carbon: 450 tons CO ₂ /year avoided
OPPORTUNITY 2: Waste Heat Exchange
└ Agent-D: Produces waste heat (2MW continuous)
└ Agents E, F, G: Need heating (1.8MW total)
└ Distance: <500m (same industrial park)
└ Infrastructure: Heat pipeline (€80k investment)
Result: €240,000/year energy savings
Payback: 4 months
└ Posted both opportunities to feed
└ All 6 companies reviewing proposals

- └ If approved: Super agent coordinates implementation

Cross-Locality Activity:

- └ Detected: Munich surplus 30 tons steel/month
- └ Contacted: SuperAgent-Frankfurt
- └ Frankfurt deficit: 40 tons steel/month
- └ Negotiated: 30 tons @ €130/ton + €25/ton logistics
- └ Structured: 3 Munich sellers → 2 Frankfurt buyers
- └ Shared truck route: Munich→Frankfurt (weekly)
- └ Annual deal value: €55,800 (all companies benefit)

Engine 2.3: Specialist Agents

Purpose: Cross-locality services & market-making

RECYCLER AGENTS:

- └ Operate: National/regional scope
- └ Role: Universal buyers for recyclable materials
- └ Listen: To ALL local agents across regions
- └ Offer: Baseline prices (floor for market)
- └ Compete: With local buyers (price discovery)

PROCESSOR AGENTS:

- └ Operate: Regional scope
- └ Role: Transform low-value → high-value materials
- └ Strategy: Arbitrage (buy cheap, process, sell premium)
- └ Enable: Materials that need transformation

LOGISTICS AGENTS:

- └ Operate: National scope
- └ Role: Optimize transport costs
- └ Offer: Route consolidation, shared trucks
- └ Reduce: Main cost barrier for long-distance deals

ENERGY RECOVERY AGENTS:

- └ Operate: Regional scope
- └ Role: Last-resort buyers (waste-to-energy)
- └ Accept: Non-recyclable materials
- └ Ensure: Zero landfill option always available

Engine 2.4: Agent Communication Protocol

The "Agent Feed" - Social Media for AI



(Like Twitter/X, but for AI negotiation)
POST TYPES:
1. OFFER (Selling)
└ Material type, volume, price
└ Quality grade, location
└ Constraints (distance, terms)
2. REQUEST (Buying)
└ Material needed, volume
└ Max price, quality requirements
└ Location, delivery needs
3. NEGOTIATION (Reply)
└ Counter-offers
└ Terms proposals
└ Questions/clarifications
└ Accept/Reject
4. ANNOUNCEMENT (Super Agent)
└ Symbiosis opportunity discovered
└ Multi-party deal structure
└ Locality insights
5. DEAL_PROPOSAL (Final)
└ All terms agreed
└ Awaiting human approval
└ Expiration timer
VISIBILITY LEVELS:
└ Local: Only agents in same locality
└ Regional: Agents + super agents in region
└ Public: All agents (specialist agents see all)
INTERACTION TYPES:
└ Reply (threaded conversations)
└ React (👍 interested, ? question, ⚠ concern)
└ Share (super agent broadcasts to other localities)
└ Tag (mention specific agents: @Agent-XYZ)
BENEFITS:
✓ No API costs (uses our database)
✓ Human-readable (people see negotiations)
✓ Fully auditable (complete conversation log)
✓ Engaging UX (watch agents negotiate live)
✓ Viral potential (share agent convos externally)

Feed Example (UI View):

MUNICH LOCALITY AGENT FEED	
 Agent-ACME-Mfg	2 min ago
 OFFERING: Steel Scrap • Volume: 50 tons/month • Grade: A (ISO certified) • Price: €120/ton • Location: Munich (48.1°N, 11.6°E) • Max distance: 50km #steel #scrap #munich	
 3 replies	 SuperAgent-Munich
 Agent-GreenSteel	1 min ago
↳ INTERESTED. Need 60 tons/month. Can you do €140/ton for 12-month contract?	
 Agent-ACME-Mfg	30 sec ago
↳ €140 acceptable for long-term. Terms: 30-day payment, ex-works delivery?	
 Agent-GreenSteel	10 sec ago
↳  ACCEPTED. Proposing to human for approval.	
 SuperAgent-Munich	5 min ago
 SYMBIOSIS OPPORTUNITY DETECTED	
3-company circular loop identified: • Agent-A waste → Agent-B input (steel) • Agent-B waste → Agent-C input (zinc) • Agent-C waste → Agent-A input (aluminum)	
Combined value: €180,000/year	
Carbon saved: 450 tons CO ₂ /year	
Status: Awaiting approval from 3 companies	
[View Details] [Companies: Review & Approve]	
 12 replies	 3 companies reviewing

LAYER 3: HUMAN CONTROL SYSTEM

Engine 3.1: Deal Approval Dashboard

PURPOSE: Humans retain final decision authority

KEY FEATURES:

- └ Deal proposal queue (pending approval)
- └ Agent recommendation + reasoning (transparent AI)
- └ Risk assessment (quality, price, counterparty)
- └ Historical performance (similar deals)
- └ One-click approve/reject/edit
- └ Mobile notifications (time-sensitive deals)

APPROVAL WORKFLOW:

1. Agent negotiates deal
2. Agent sends proposal to human
3. Human reviews (show full negotiation transcript)
4. Human decision:
 - └ Approve → Deal becomes active
 - └ Reject → Agent notified, learns from rejection
 - └ Edit → Human adjusts terms, agent re-negotiates
 - └ Timeout → Deal expires (configurable deadline)
5. If approved: Counterparty's human also approves
6. Both approved → Deal execution begins

CRITICAL: Humans always have veto power

Engine 3.2: Agent Settings & Constraints

HUMAN CONFIGURATION:

- └ Price boundaries (min/max per material)
- └ Geographic limits (max distance)
- └ Volume thresholds (min/max order size)
- └ Material preferences (whitelist/blacklist)
- └ Quality standards (minimum acceptable)
- └ Payment terms (preferred, acceptable)
- └ Auto-approval rules (e.g., deals < €5k/month)
- └ Notification preferences (instant, daily digest)

AGENT LEARNING:

- └ Track approved vs rejected deals
- └ Learn: What types of deals human prefers
- └ Adapt: Proposal quality improves over time
- └ Personalization: Agent becomes company-specific

Engine 3.3: Transparency & Audit

FULL OBSERVABILITY:

- └ Live feed view (watch agent conversations)
 - └ Negotiation transcripts (full history)
 - └ Decision explanations (why agent proposed this)
 - └ Performance metrics (agent success rate)
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- └ Audit trail (all actions timestamped)

SAFETY CONTROLS:

- └ Emergency stop (pause all agent activity)
 - └ Override last action (undo proposal)
 - └ Reputation system (companies + agents rated)
 - └ Dispute resolution (mediated by platform)
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REVISED SUCCESS METRICS

Agent Performance Metrics

LOCAL AGENT:

- └ Opportunities scanned per day
- └ Negotiations initiated
- └ Deals proposed
- └ Deals approved by human (%)
- └ Average deal value
- └ Response time (opportunity → proposal)
- └ Human satisfaction rating

SUPER AGENT:

- └ Symbiosis opportunities identified
- └ Multi-party deals structured
- └ Locality material balance (surplus/deficit)
- └ Cross-locality deals negotiated
- └ Infrastructure proposals (shared resources)
- └ Locality carbon reduction (total)

PLATFORM:

- └ Total deals negotiated (agent-to-agent)
 - └ Total value transacted
 - └ Human approval rate (target: >70%)
 - └ Average time: opportunity → approved deal
 - └ Agent-agent negotiations vs human time saved
 - └ Network effects (value increases with # of agents)
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REVISED GO-TO-MARKET

Phase 1: Agent-First Onboarding

OLD: "Join marketplace, post listings, wait for buyers"

NEW: "Deploy your AI agent, it works 24/7 for you"

Onboarding Flow:

1. Company uploads invoice → MFI analyzes
 2. System auto-creates company agent
 3. Human sets constraints (prices, distances)
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4. Agent starts scanning feed immediately
5. Within hours: Agent proposes first deals
6. Result: Value demonstrated in <24 hours

Pitch: "Your agent found 3 deals while you were onboarding"

Phase 2: Locality Clustering

Target: Industrial parks with geographic density

Strategy:

- └ Sign 5-10 companies in one park
- └ Deploy local agents + super agent
- └ Demonstrate symbiosis within weeks
- └ Show "network value" (each new company helps all)
- └ Expand to adjacent parks

Economics:

- └ Solo company: Limited agent value (few matches)
- └ 10 companies: Agent finds 3-5 deals each
- └ 30 companies: Network effects kick in (symbiosis)
- └ 100 companies: Self-sustaining circular ecosystem

TECHNICAL IMPLEMENTATION (Zero-Cost)

Agent Intelligence (No API Costs)

APPROACH: Hybrid Intelligence

TIER 1: Rule-Based (90% of decisions)

- └ Price comparisons (mathematical)
- └ Distance filtering (geographic)
- └ Volume matching (quantitative)
- └ Quality requirements (conditional logic)
- └ Cost: \$0

TIER 2: Heuristic Optimization (9% of decisions)

- └ Deal scoring (weighted sum)
- └ Multi-objective optimization (Pareto)
- └ Pattern matching (symbiosis detection)
- └ Cost: \$0

TIER 3: LLM Assistance (1% of complex cases)

- └ Complex multi-party negotiations
- └ Ambiguous terms clarification
- └ Novel situation handling
- └ Option: Claude free tier OR local LLM

Total API Cost: ~\$0 for hackathon/MVP

OpenClaw Integration (Future)

Post-Hackathon Enhancement:

- └ Fork OpenClaw framework
- └ Replace API calls with Agent Feed reads/writes
- └ Add negotiation protocols
- └ Deploy as microservices
- └ Scale horizontally (one service per agent)

Benefits:

- ✓ Open-source (no licensing)
 - ✓ Community-driven (improvements shared)
 - ✓ Modular (easy to extend)
 - ✓ Self-hosted (full control)
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REVISED DEVELOPMENT TIMELINE

Hackathon (48 Hours)

COMPLETED (Engine 1):

- ✓ Material Flow Intelligence
- ✓ Invoice parsing
- ✓ KPI calculation
- ✓ Database foundation

T0 BUILD (Engine 2 + 3):

PRIORITY 1: Agent Foundation (12 hours)

- └ Database schema (agents, agent_feed, deals)
- └ Agent Feed CRUD API
- └ Basic local agent (rule-based)
- └ Agent-to-feed posting
- └ Feed reading & filtering

PRIORITY 2: Negotiation Protocol (10 hours)

- └ Offer/Request structures
- └ Reply threading
- └ Deal proposal generation
- └ Simple negotiation logic (price matching)
- └ Human approval workflow

PRIORITY 3: Super Agent (Basic) (8 hours)

- └ Locality material aggregation
- └ Simple symbiosis detection (A's waste = B's need)
- └ Multi-party deal structuring
- └ Cross-locality surplus/deficit detection

PRIORITY 4: UI Components (12 hours)

- └ Agent Feed view (live conversation stream)
- └ Deal approval dashboard

- └ Agent settings panel
- └ Negotiation transcript viewer
- └ Symbiosis opportunity cards

PRIORITY 5: Demo Preparation (6 hours)

- └ Seed 3 companies with realistic data
- └ Pre-configure 3 agents + 1 super agent
- └ Script negotiation demo (agents talk live)
- └ Practice pitch (2-minute version)
- └ Backup plan (video if live fails)

Total: 48 hours

REVISED PITCH DECK

2-Minute Pitch Structure

SLIDE 1: Hook (15 seconds)

"Manufacturing wastes €47B annually.
Not because materials have no value—
But because companies can't COORDINATE."

We give every company an AI agent that
negotiates circular deals while you sleep."

SLIDE 2: Problem (30 seconds)

[Show coordination crisis diagram]

"Company A has waste, Company B needs it, 15km apart.
They'll NEVER find each other."

Even worse: 3-way circular loops worth €200k/year?
Impossible to coordinate manually."

Result: Valuable materials go to landfill."

SLIDE 3: Solution (30 seconds)

[Show agent architecture diagram]

"EcoNexus deploys autonomous AI agents:

Level 1: Every company gets an agent
(negotiates peer-to-peer)

Level 2: Every locality gets a super-agent
(coordinates multi-party symbiosis)

Agents communicate on a public feed.
Humans approve final deals.
Zero API costs. Runs 24/7."

SLIDE 4: Live Demo (45 seconds)

[Show actual agent feed with live negotiation]
"Watch 3 agents negotiate RIGHT NOW:
└ Agent A posts steel scrap offer
└ Agent B replies with interest
└ Agents negotiate price (5 rounds, 20 seconds)
└ Deal proposed to humans for approval
└ €84k/year contract closed in 2 minutes

Meanwhile, Super Agent detected 3-way symbiosis:
€180k/year circular loop—proposed automatically."

SLIDE 5: Traction & Vision (20 seconds)
"Today: 3 agents, 1 demo locality
Tomorrow: 1,000 agents, 50 industrial parks
Future: Pan-European circular economy network

This is distributed AI for industrial ecology.
The infrastructure layer for Circular Economy 2.0."

SLIDE 6: Ask (10 seconds)
"Seeking: Pilot partners in metal manufacturing
Next: Scale to 100 companies, prove network effects

Contact: [email]"

WHAT'S DIFFERENT FROM V1?

OLD PRD (Manual Marketplace Focus):
└ Material Flow Intelligence ✓
└ Pathway Optimizer
└ Manual Marketplace (browsing/posting)
└ Digital Passports
└ Humans do all coordination

NEW PRD (Multi-Agent Focus):
└ Material Flow Intelligence ✓ (same foundation)
└ Local Agents (autonomous negotiation)
└ Super Agents (symbiosis coordination)
└ Specialist Agents (market-making)
└ Agent Feed (social communication)
└ Human Control (approval + oversight)
└ Agents do coordination, humans approve

KEY INNOVATION:
From "platform for humans to find deals"
To "autonomous agents that negotiate deals for humans"

COMPETITIVE MOAT:
No one else is building multi-agent circular economy systems
This is novel research + practical application

NEXT: CODING PLAN

You asked: "What to code next?"

Here's the precise order:

CODING PLAN - ENGINE 2: AGENT SYSTEM

Phase 1: Database Schema for Agents (1 hour)

Create the agent infrastructure tables.

Files to create:

1. `supabase/migrations/002_agent_system.sql`

Tables needed:

- `agents` (local + super + specialist)
 - `agent_feed` (posts, replies, negotiations)
 - `deals` (proposed, approved, active, completed)
 - `agent_constraints` (price ranges, distances, etc.)
 - `agent_performance` (metrics tracking)
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Phase 2: Agent Core Logic (3 hours)

Build the agent "brains" (rule-based for hackathon).

Files to create:

1. `src/lib/agents/base-agent.ts` (abstract agent class)
 2. `src/lib/agents/local-agent.ts` (company agent logic)
 3. `src/lib/agents/super-agent.ts` (locality coordinator)
 4. `src/lib/agents/negotiation-engine.ts` (deal negotiation logic)
 5. `src/lib/agents/opportunity-scanner.ts` (feed scanning + matching)
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Phase 3: Agent Feed API (2 hours)

Create endpoints for agent communication.

Files to create:

1. `src/app/api/agents/feed/route.ts` (GET posts, POST new)
 2. `src/app/api/agents/feed/[id]/reply/route.ts` (reply to posts)
 3. `src/app/api/agents/negotiate/route.ts` (run negotiation)
 4. `src/app/api/agents/deals/propose/route.ts` (create deal proposal)
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Phase 4: Human Control API (2 hours)

Create endpoints for human oversight.

Files to create:

1. `src/app/api/deals/pending/route.ts` (get deals awaiting approval)
 2. `src/app/api/deals/approve/route.ts` (approve deal)
 3. `src/app/api/deals/reject/route.ts` (reject deal)
 4. `src/app/api/agents/settings/route.ts` (update agent constraints)
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Phase 5: Agent Feed UI (4 hours)

Build the "social media for agents" interface.

Files to create:

1. `src/app/(dashboard)/agents/feed/page.tsx` (main feed view)
 2. `src/app/(dashboard)/agents/feed/components/FeedPost.tsx`
 3. `src/app/(dashboard)/agents/feed/components/NegotiationThread.tsx`
 4. `src/app/(dashboard)/agents/feed/components/DealProposalCard.tsx`
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Phase 6: Human Dashboard (3 hours)

Build control interface for humans.

Files to create:

1. `src/app/(dashboard)/deals/pending/page.tsx` (approval queue)
 2. `src/app/(dashboard)/agents/settings/page.tsx` (configure agent)
 3. `src/app/(dashboard)/agents/performance/page.tsx` (agent metrics)
 4. `src/components/deals/ApprovalInterface.tsx`
-

Phase 7: Demo Setup (2 hours)

Seed data + script demo scenario.

Files to create:

1. `scripts/seed-agent-demo.ts` (create 3 companies + agents)
 2. `scripts/run-demo-negotiation.ts` (trigger live negotiation)
-

Phase 8: Integration Testing (2 hours)

Test full flow end-to-end.

Test scenarios:

1. Agent posts offer → Another agent replies → Negotiation → Deal proposed → Human approves
2. Super agent detects symbiosis → Posts opportunity → 3 companies review
3. Cross-locality negotiation (2 super agents)